

Git and GitHub

Submitted to Vikul_Sir

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L3 - Using Local and Remote git repositories demonstrate git pull and git fetch. Compare the differences

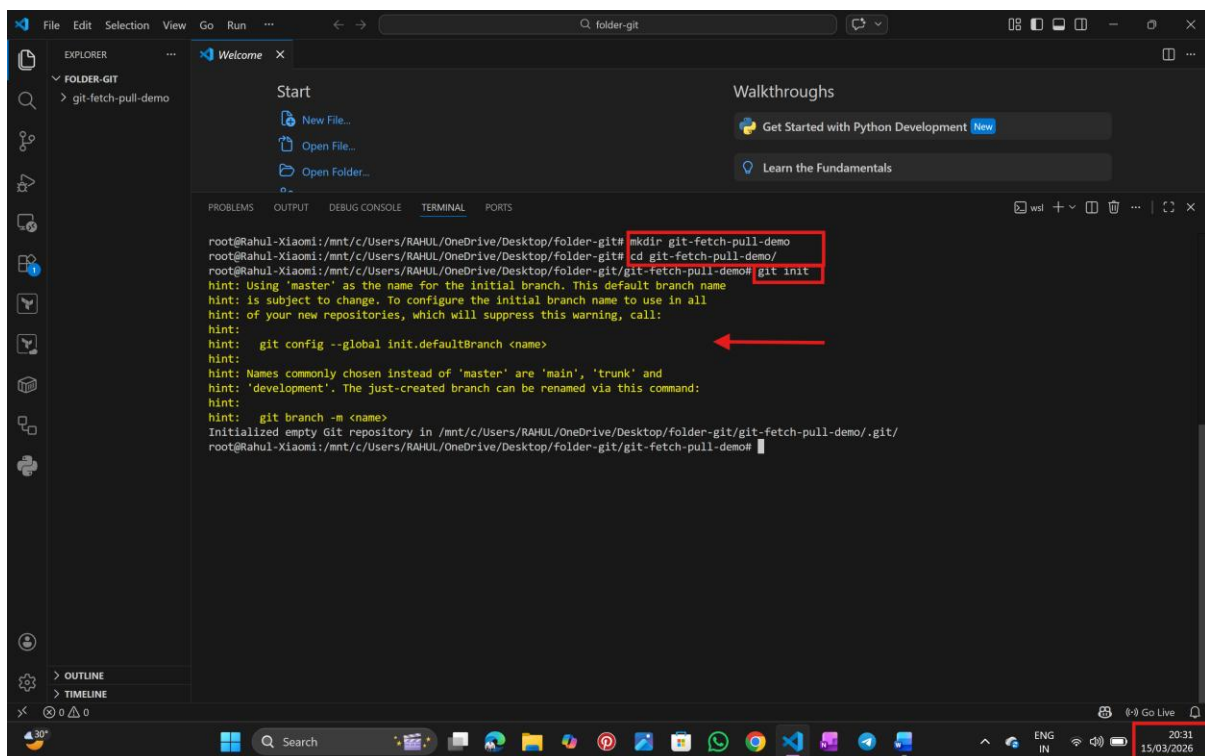
Step :1 Create a Local Repository

→open the VSC-visual studio code

→create a directory, use command – mkdir git-fetch-pull-demo

→change to the directory, use command – cd git -fetch-pull-demo

→initialization the git, use command – git init



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the following commands and output:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git# mkdir git-fetch-pull-demo
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git# cd git-fetch-pull-demo/
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git init
```

The output of the `git init` command is shown below:

```
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo/.git/
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#
```

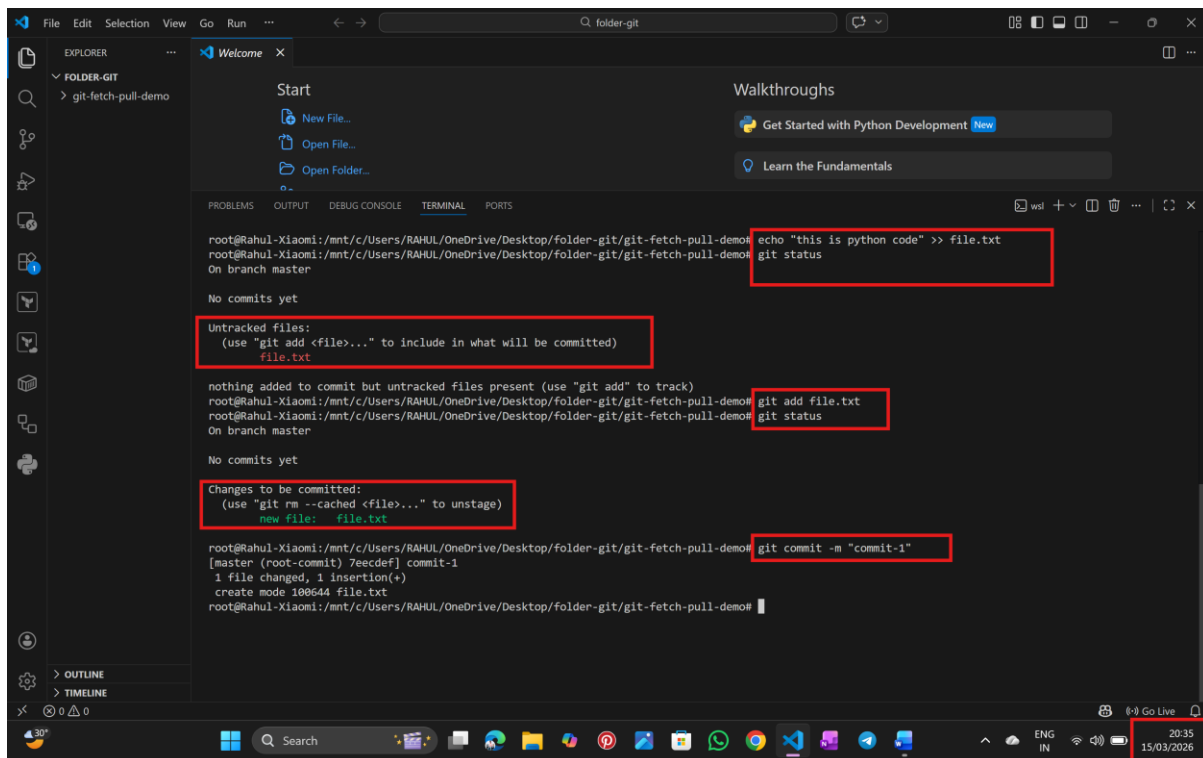
A red box highlights the commands `mkdir git-fetch-pull-demo`, `cd git-fetch-pull-demo/`, and `git init`. A red arrow points to the `git init` command. The system tray at the bottom right shows the date and time as 20:31 on 15/03/2026.

Step :2 create a file

→create a file with content it, use command – echo “content” >> <filename>

→next use command – git add <file>

→and commit, use command – git commit -m “commit message”



The screenshot shows a terminal window in VS Code with the following commands and output:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# echo "this is python code" >> file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file.txt

nothing added to commit but untracked files present (use "git add" to track)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git add file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git status
On branch master

No commits yet

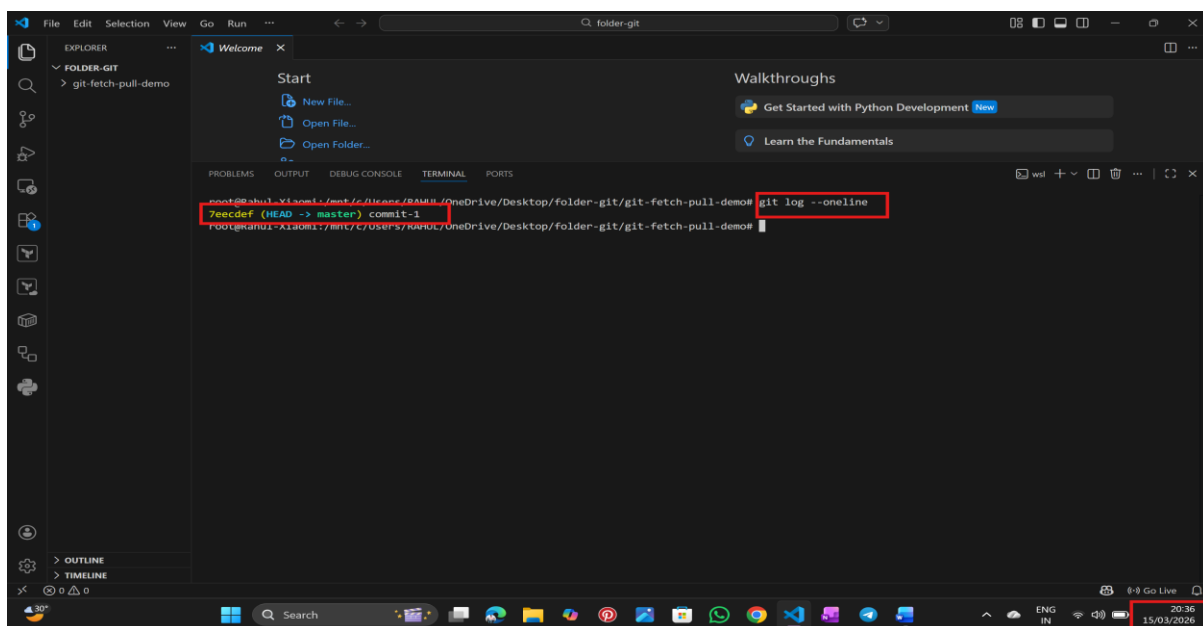
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file.txt

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git commit -m "commit-1"
[master (root-commit) 7eeecdef] commit-1
1 file changed, 1 insertion(+)
create mode 100644 file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#
```

The terminal window is titled "folder-git" and shows the output of the commands. The file "file.txt" is created and added to the commit. The commit message is "commit-1". The terminal window is also showing the output of the "git status" command, which indicates that the file is untracked and has been added to the commit.

Step :3 show the git commit history

→use command – git log –oneline



The screenshot shows a terminal window in VS Code with the following command and output:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git log --oneline
7eeecdef (HEAD -> master) commit-1
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#
```

The terminal window is titled "folder-git" and shows the output of the "git log --oneline" command. The output shows the commit hash "7eeecdef" and the commit message "commit-1". The terminal window is also showing the output of the "git log --oneline" command, which indicates that the file is untracked and has been added to the commit.

Step :4 create Remote Repository

→go to GitHub

→click new repo, with new name

→and copy the HTTP URL of the repo

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#). Required fields are marked with an asterisk (*).

1 General

Owner *  rahulnalathati-21

Repository name *
 git-fetch-pull-demo is available.

Great repository names are short and memorable. How about [ideal-telegram?](#)

Description
 0 / 350 characters

2 Configuration

Choose visibility *
 Choose who can see and commit to this repository

Add README
 READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore
 .gitignore tells git which files not to track. [About ignoring files](#)

Add license
 Licenses explain how others can use your code. [About licenses](#)

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git-fetch-pull-demo (Public)

Set up GitHub Copilot
 Use GitHub's AI pair programmer to autocomplete suggestions as you code.

Add collaborators to this repository
 Search for people using their GitHub username or email address.

Quick setup — if you've done this kind of thing before
 or <https://github.com/rahulnalathati-21/git-fetch-pull-demo.git>
 Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# git-fetch-pull-demo" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/rahulnalathati-21/git-fetch-pull-demo.git
git push -u origin main
```

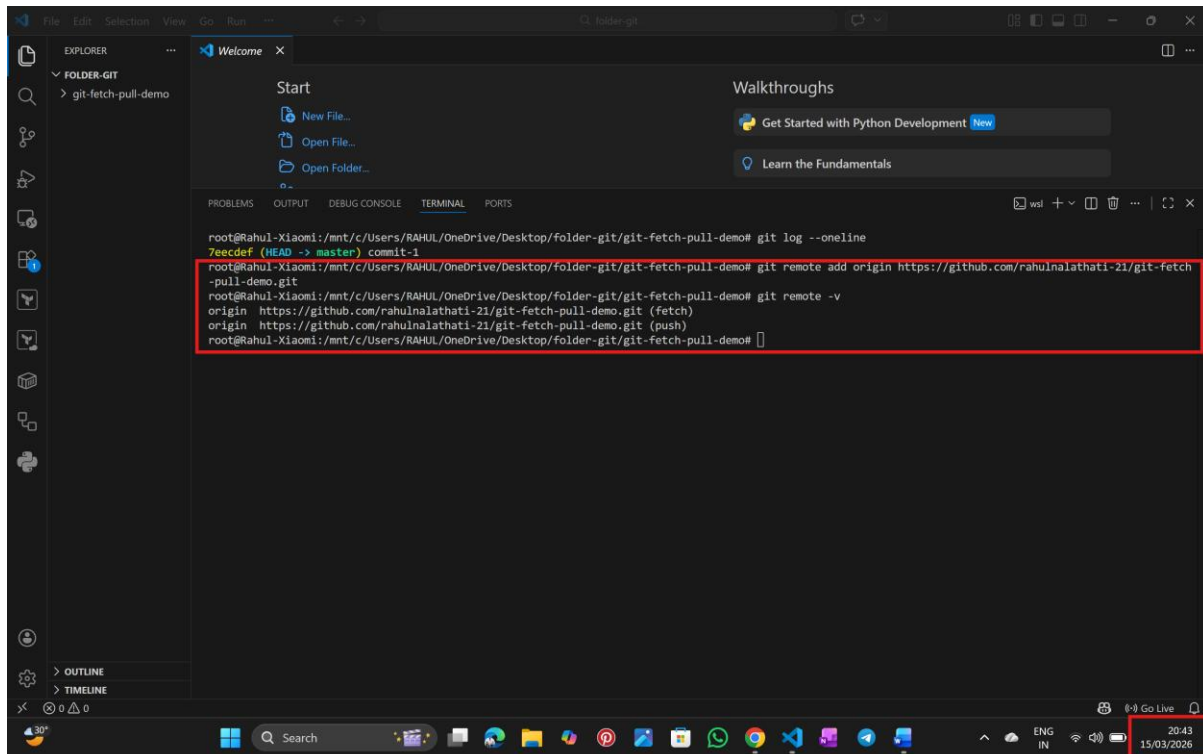
...or push an existing repository from the command line

```
git remote add origin https://github.com/rahulnalathati-21/git-fetch-pull-demo.git
git branch -M main
git push -u origin main
```

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Step :5 Connect Local Repo to Remote Repo

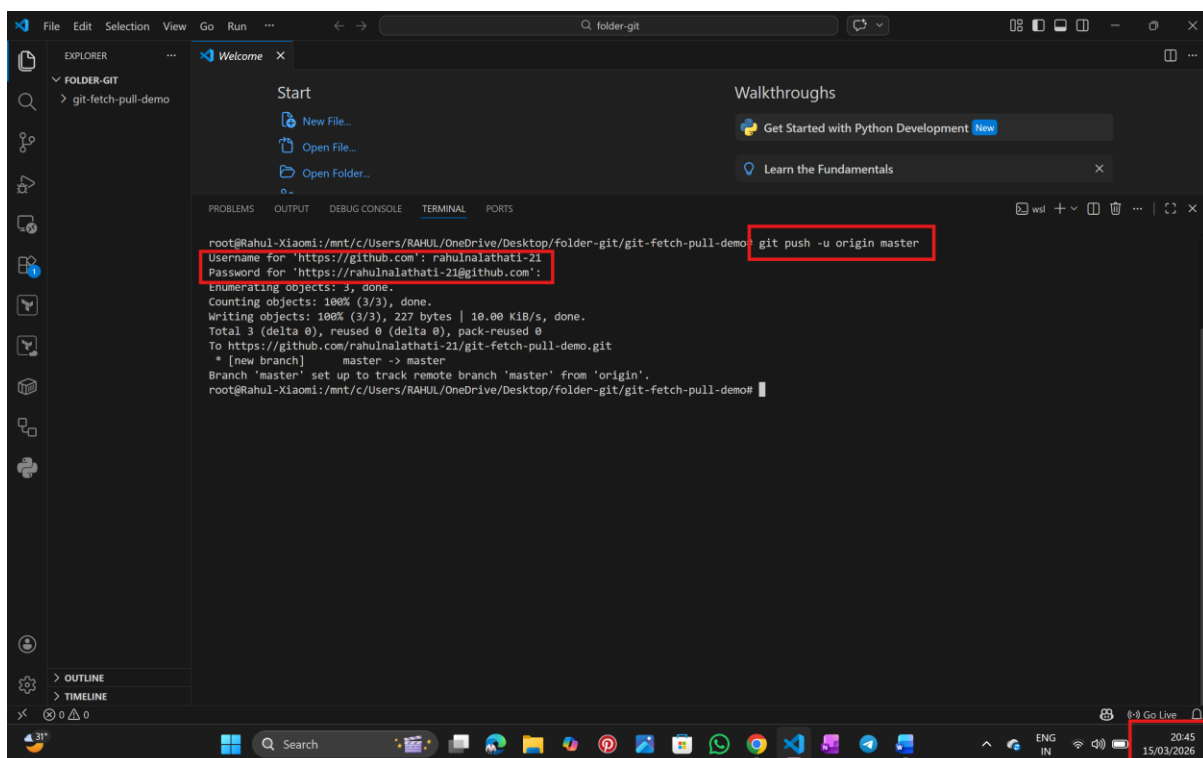
→clone the repo to local, use command – git clone <URL>



```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git log --oneline
7e2cdef (HEAD -> master) commit-1
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git remote add origin https://github.com/rahulnalathati-21/git-fetch-pull-demo.git
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git remote -v
origin https://github.com/rahulnalathati-21/git-fetch-pull-demo.git (fetch)
origin https://github.com/rahulnalathati-21/git-fetch-pull-demo.git (push)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#
```

Step :6 Push Local Repository to Remote

→use command -- git push -u origin master

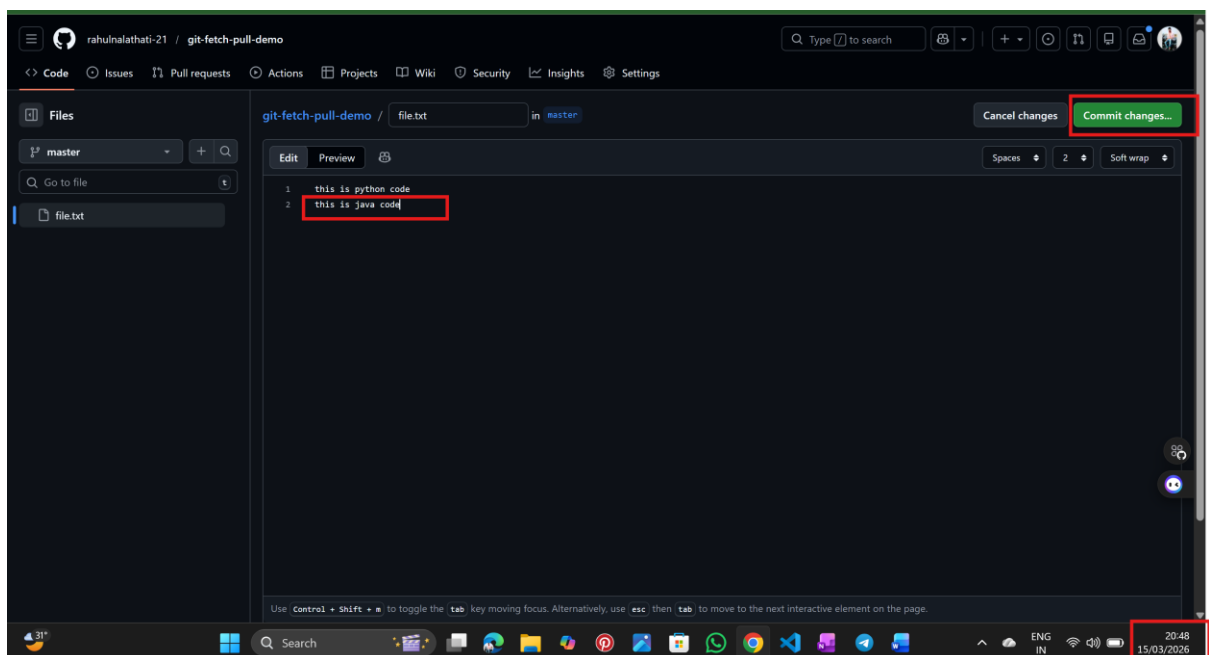
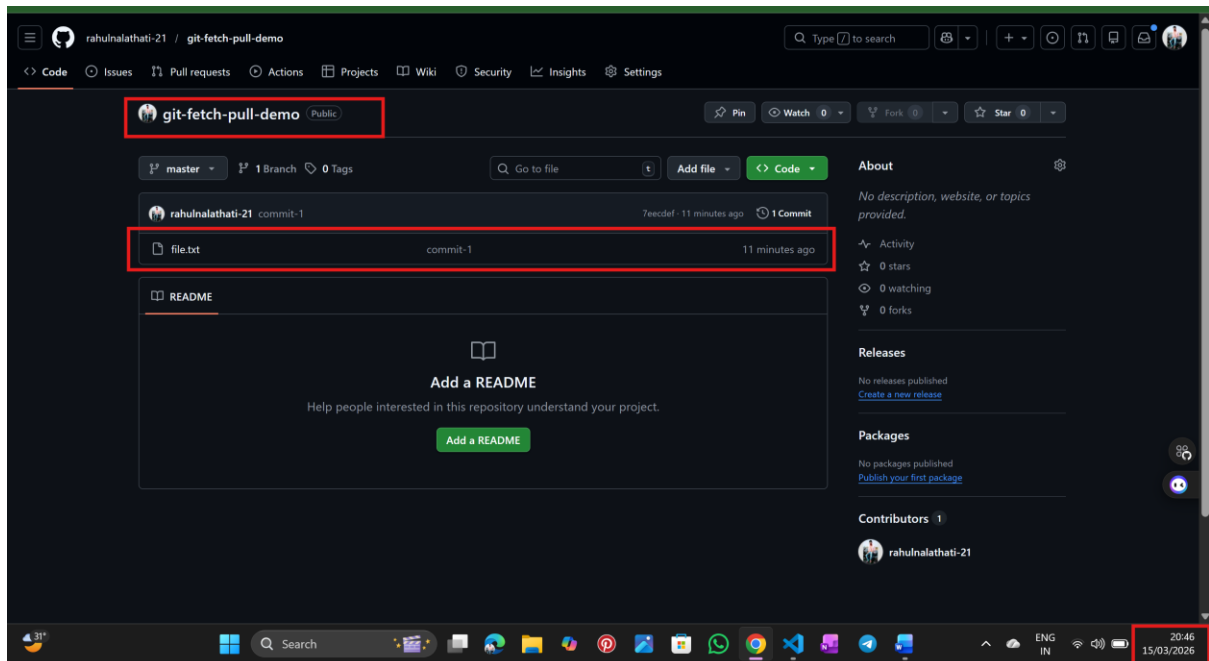


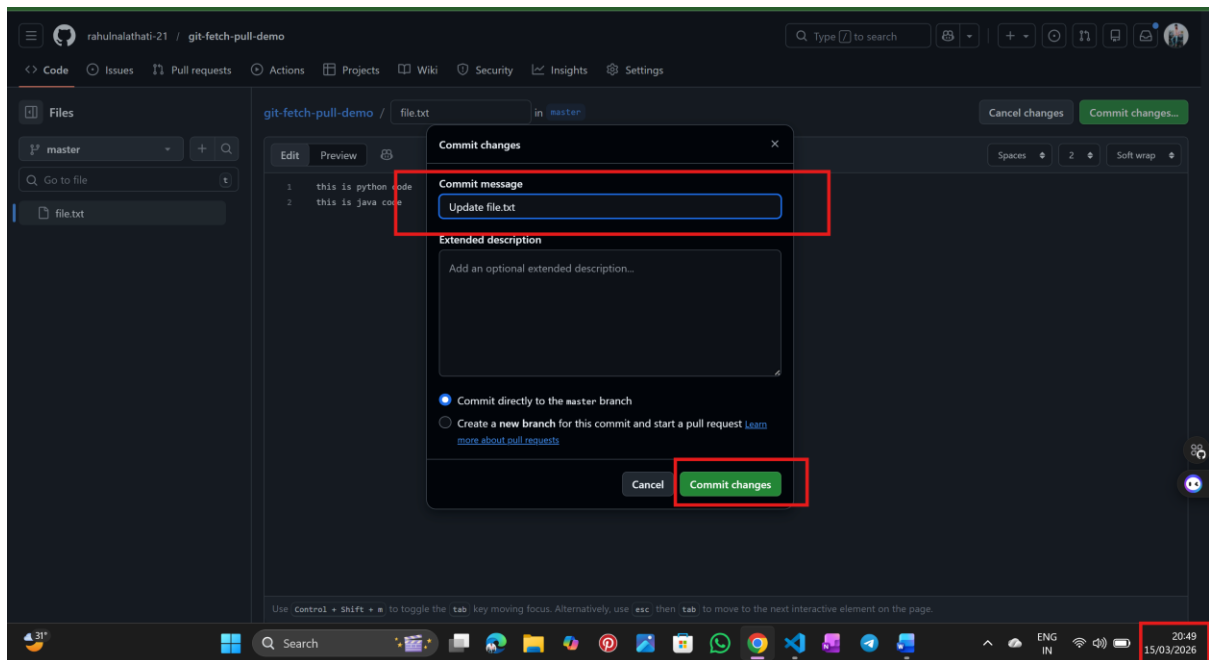
```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git push -u origin master
Username for 'https://github.com': rahulnalathati-21
Password for 'https://github.com': 
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 227 bytes | 10.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/rahulnalathati-21/git-fetch-pull-demo.git
 * [new branch] master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#
```

Step :7 now make change in remote repository manually

→go to git hub and edit the file

→and commit change their itself



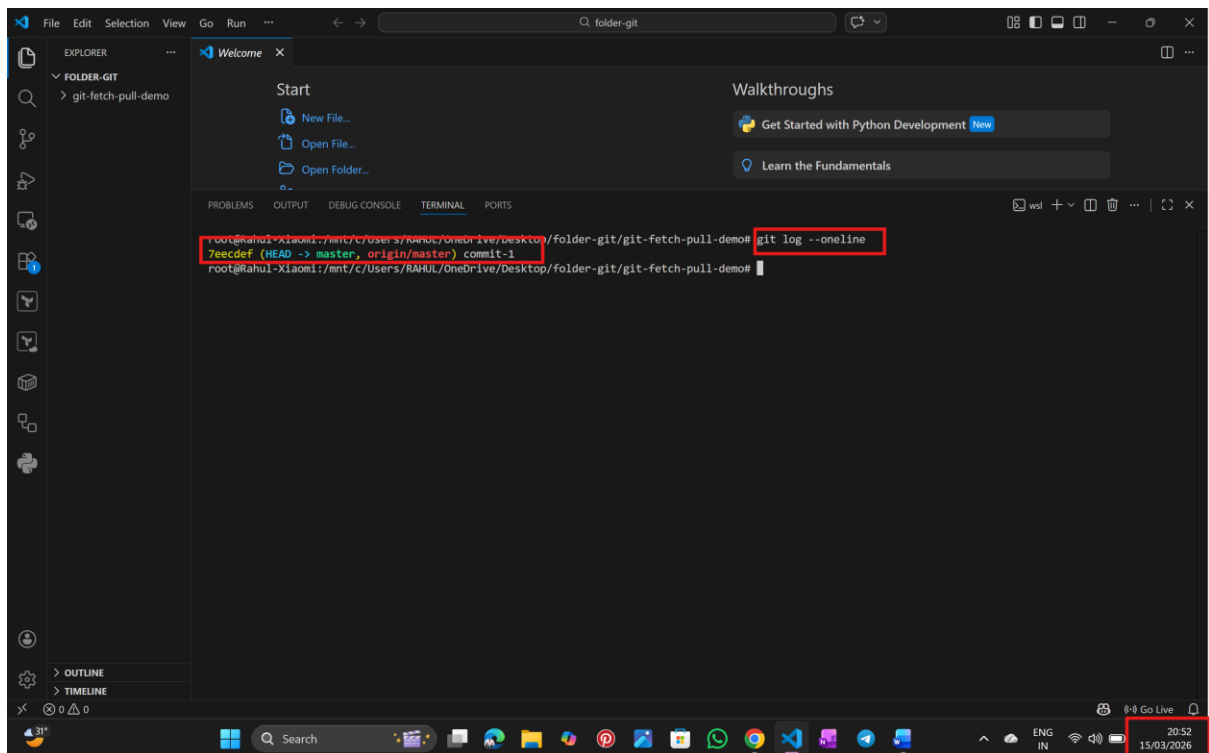


Step :8 now got to terminal

→use command -- git fetch origin

→git log origin/master --oneline

→cat file.txt - You will see **remote change is not yet in the file.**



This screenshot shows a VS Code terminal window with the following content:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git log --oneline
7eecdef (HEAD -> master, origin/master) commit-1
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git fetch origin
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 918 bytes | 7.00 KiB/s, done.
From https://github.com/rahulnalathati-21/git-fetch-pull-demo
  7eecdef..8293482  master    -> origin/master
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git log origin/master
commit 829348208cdfcb6f4c46323166fe8e433fa4ec44 (origin/master)
Author: rahulnalathati-21 <rahulsiddharth2.0@gmail.com>
Date:   Sun Mar 15 20:50:47 2026 +0530

    Update file.txt

commit 7eecdefe1e38c581303c206ab19ca081048e4ba (HEAD -> master)
Author: Rahul <rahulsiddharth2.0@gmail.com>
Date:   Sun Mar 15 20:35:03 2026 +0530

    commit-1
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#
```

The terminal output shows the execution of `git log --oneline`, `git fetch origin`, and `git log origin/master`. The `git fetch origin` command is highlighted with a red box. The system tray at the bottom right shows the time as 20:54 on 15/03/2026, also highlighted with a red box.

This screenshot shows a VS Code terminal window with the following content:

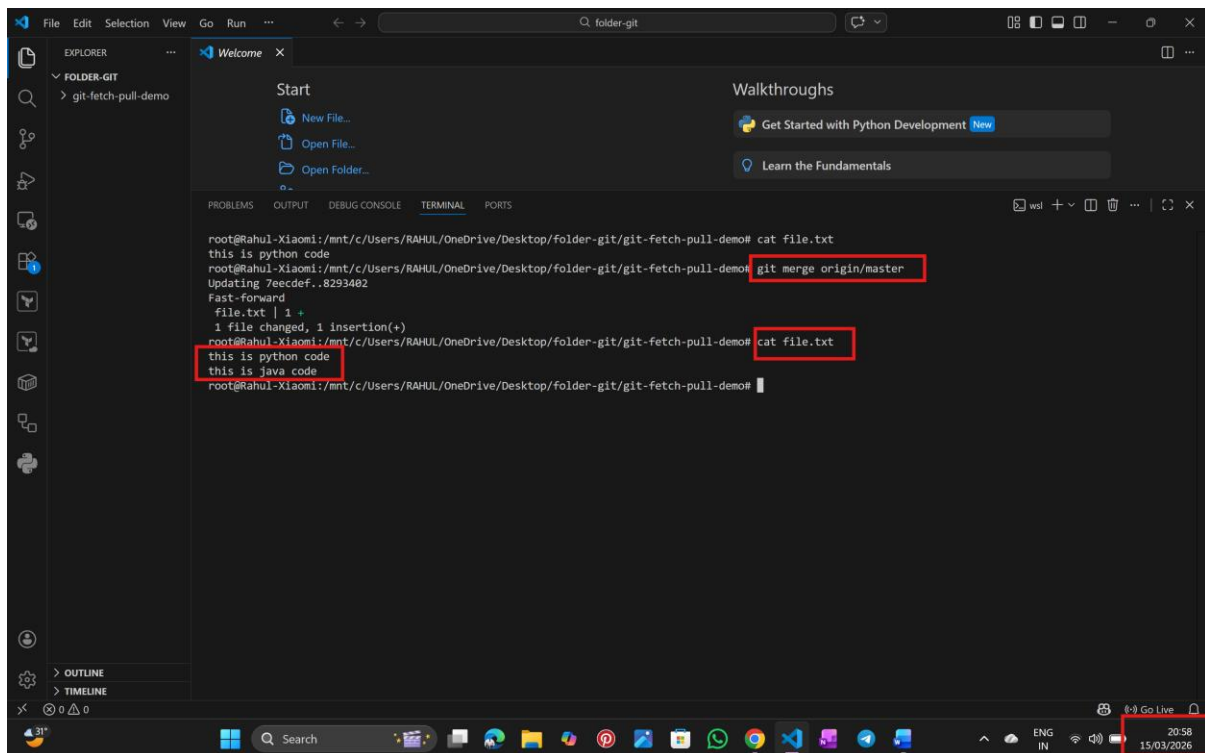
```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# cat file.txt
this is python code
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#
```

The terminal output shows the execution of `cat file.txt`, which displays the content of the file. The `cat file.txt` command is highlighted with a red box. The system tray at the bottom right shows the time as 20:56 on 15/03/2026, also highlighted with a red box.

Step: 9 Merge After Fetch

→git merge origin/master

→cat file.txt - Now the remote change appears in the file.

A screenshot of a Visual Studio Code editor window. The Explorer sidebar on the left shows a folder named 'FOLDER-GIT' with a subfolder 'git-fetch-pull-demo'. The main editor area displays a terminal window with the following text: 'root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# cat file.txt', 'this is python code', 'root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# git merge origin/master', 'Updating 7ee2def..8293402', 'Fast-forward', 'file.txt | 1 +', '1 file changed, 1 insertion(+)', 'root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo# cat file.txt', 'this is python code', 'this is java code', and 'root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-fetch-pull-demo#'. The commands 'git merge origin/master' and 'cat file.txt' are highlighted with red boxes. The bottom status bar shows '20:58 15/03/2026'.

Step: 10 Comparison between git fetch and git pull

git fetch

→Safe operation

→Only downloads changes

→Manual merge required

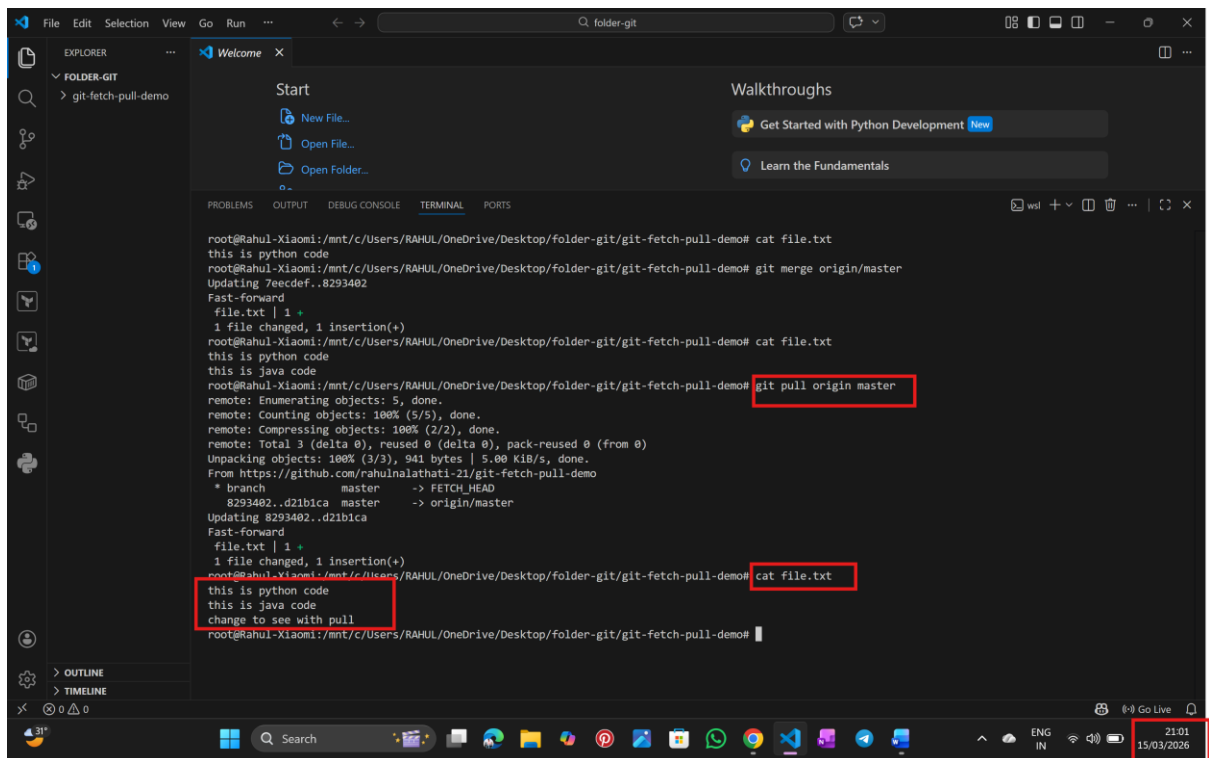
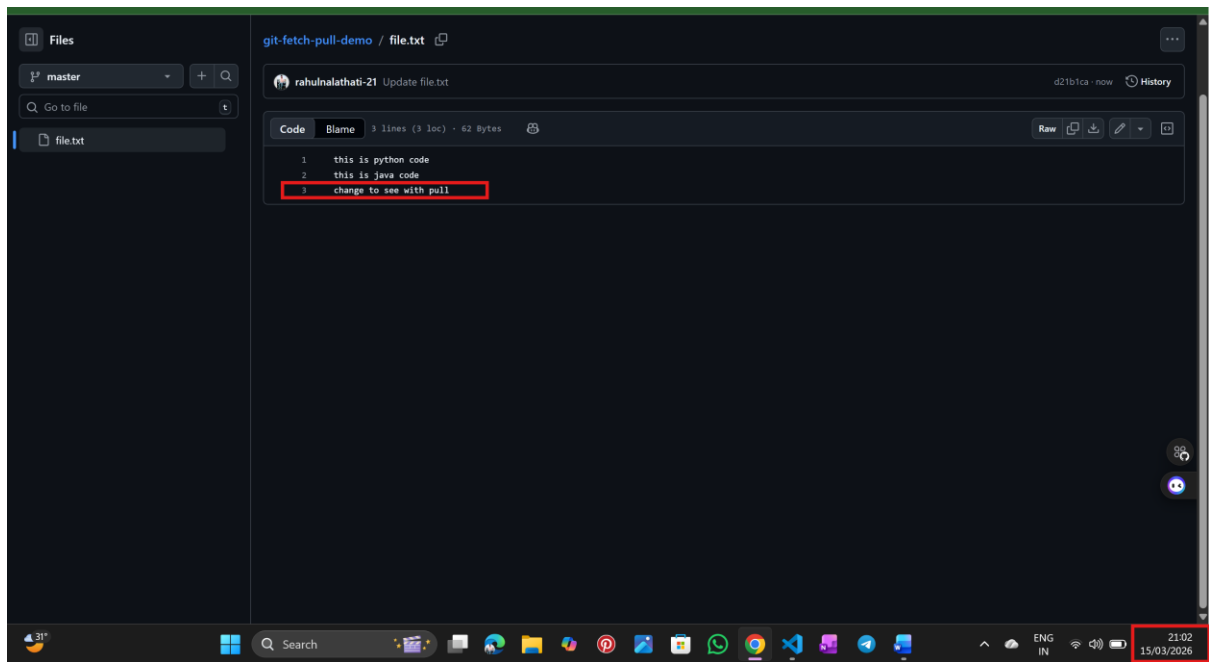
→Used when you want to review changes first

git pull

→Combines fetch + merge

→Automatically updates local repository

→Faster but may cause merge conflicts



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